

HW-4

$$[1] \quad f = 1500 \text{ Hz} ; \quad \bar{\omega} = 1500 \times 2\pi = 9424.78 \text{ rad/s.}$$

$$\omega = 2E3 \text{ lb} \quad TR = 0.1$$

$$TR = \frac{1}{|1 - \beta^2|} = \frac{1}{|1 - (\frac{\bar{\omega}}{\omega_n})^2|} = 0.1$$

$$|1 - (\frac{\bar{\omega}}{\omega_n})^2| = 10 \quad (\frac{\bar{\omega}}{\omega_n})^2 = 11$$

$$\frac{\bar{\omega}}{\omega_n} = \sqrt{11}$$

$$\omega_n = 2841.67 \text{ rad/s.}$$

$$\omega_n^2 = \frac{k}{m}$$

$$(2841.67)^2 = \frac{k}{\frac{2000}{386}}$$

$$\Rightarrow k = 41,839,836.21 \text{ lb/in.}$$

$$[2] \quad \omega = 500 \text{ kips; } f = 50 \text{ kips; } T_n = 0.25 \text{ sec, } P_n = 100 \text{ kips;}$$

$$T_0 = 0.3 \text{ sec; } \omega_n = 8\pi; \quad \omega_D = \omega_n \sqrt{1 - \xi^2}$$

$$\Rightarrow \frac{20\pi}{3} = 8\pi \sqrt{1 - \xi^2}$$

$$\Rightarrow \xi = 0.5527$$

$$\beta = \frac{\omega_D}{\omega_n} = \frac{20/3}{24} = 0.833$$

$$\text{Now, } (U_{st})_0 = \frac{P_0}{k}$$

$$\text{where, } P_0 = 100 \text{ kips}$$

$$k = \omega_n^2 m = \left(\frac{2\pi}{T_n}\right)^2 \frac{W}{g} = (8\pi)^2 \frac{500}{386} = 818 \text{ kips/in}$$

The

$$\text{Thus, } (u_{st})_0 = \frac{100}{818} = 0.1222 \text{ in}$$

$$\text{and, } u_0 = 2.52 * (0.1222) = 0.308 \text{ in}$$